

Week 4 Task 3 — EAL Port Init Test Report

kyocera_pmd_init | LX2160A (DPAA2) | April 2026

Test Environment

Item	Detail
Board	NXP LA1224-RDB RevC (LX2160A)
OS	Ubuntu 20.04, Linux 5.15.71-rt51
DPDK	22.11 (static libs)
App	kyocera_pmd_init (Week 4 Task 3)
Host	Mac Studio (10GbE en0)
Cable	Cat6 UTP — 10G PORT 2 (dpmac.4)

Setup Commands

```
# Switch to DPDK mode
mode dpdk
export DPRC=dprc.2

# Clean before each run
rm -f /mnt/huge/* && rm -rf /var/run/dpdk/kpmd/

# Run application
cd /root/dpdk_workspace/week4/Task_3_EAL_Port_Init/src/
./kyocera_pmd_init -l 0-3 -n 1 --file-prefix=kpmd
```

Test 1 — Ping Flood (ARP Frames)

Mac command (after 'Polling port=0' appears):

```
ping -f -c 5000 192.168.10.11 2>/dev/null &
```

Board output:

```
KPMD: Found 1 Ethernet port(s).
KPMD: Port 0: driver=net_dpaa2 max_rxq=128 max_txq=16
KPMD: Port 0 MAC 00:04:9f:08:02:78
KPMD: Port 0 link DOWN speed 10000 Mbps duplex full
KPMD: Polling port=0 q=0 for 30 s on lcore 0...
KPMD: Done. RX mbufs=99 | ipackets delta=99 | ibytes delta=49460
      imissed delta=0 | ierrors delta=0
```

Metric	Value
RX frames	99
Missed	0

Errors	0
Frame type	ARP broadcast

Test 2 — Scapy UDP (Real UDP Frames)

Mac command (after 'Polling port=0' appears):

```
sudo python3 /tmp/send_udp.py
```

Scapy script:

```
from scapy.all import *
import time
BOARD_MAC = '00:04:9f:08:02:78'
pkt = Ether(dst=BOARD_MAC) / IP(src='192.168.10.20', dst='192.168.10.11') /
    UDP(sport=1234, dport=5678) / Raw(b'KyoceraTask3' * 100)
time.sleep(10) # wait for link
sendp(pkt, iface='en0', count=5000, inter=0.0002, verbose=False)
```

Board output:

```
KPMD: Found 1 Ethernet port(s).
KPMD: Port 0: driver=net_dpaa2 max_rxq=128 max_txq=16
KPMD: Port 0 MAC 00:04:9f:08:02:78
KPMD: Port 0 link DOWN speed 10000 Mbps duplex full
KPMD: Polling port=0 q=0 for 30 s on lcore 0...
KPMD: Done. RX mbufs=5080 | ipackets delta=5080 | ibytes delta=6250928
      imissed delta=0 | ierrors delta=0
```

Metric	Value
Sent (Mac)	5000 UDP packets
RX frames	5080
Missed	0
Errors	0
Frame type	Real IPv4/UDP

Results Summary

Test	Traffic	Sent	Received	Missed	Errors	Status
Test 1	Ping Flood	~5000 ARP	99	0	0	PASS
Test 2	Scapy UDP	5000 UDP	5080	0	0	PASS

Task 3 Checklist

Checkpoint	Result
EAL init (16 lcores detected)	PASS
DPAA2 port discovered (net_dpaa2)	PASS
Port MAC address printed	PASS
RX/TX queues configured	PASS
30s poll loop completed	PASS
RX mbufs > 0	PASS
imissed = 0	PASS
ierrors = 0	PASS

Verdict: Task 3 PASSED

Platform: NXP LA1224-RDB RevC | April 2026